Process Characterization Plan

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCP-004

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

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Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Purification Process Development	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

CPP, critical process parameter · CQA, critical quality attribute · DNA, deoxyribonucleic acid · DS, drug substance · GPP, general process parameter · HCP, host-cell protein · ICH, International Council for Harmonisation · IPC, in-process control · KPP, key process parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range · NTU, nephelometric turbidity unit · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · rcf, relative centrifugal force · SDM, scale-down model · SEC, size-exclusion chromatography.

1 Purpose and scope

This Process Characterization Plan (**PCP-004**) defines the Stage 1 (Process Design) characterization of the A-Mab harvest and clarification operation (Step 4), the unit operation that separates the antibody-containing supernatant from the cells and cell debris of the production culture and delivers a clarified feed to the Protein A capture step. The plan specifies the objectives, the process-performance measures in scope, the process parameters and ranges to be characterized, the analytical methods, the sampling plan, the analysis approach, and the acceptance and decision criteria by which each parameter will be classified.

Scope. The primary recovery train — continuous disk-stack centrifugation followed by depth and sterile (bioburden-reduction) filtration — that clarifies the 15,000-L production-bioreactor harvest, evaluated on a qualified bench-scale clarification model. The production bioreactor (Step 3) and the Protein A capture step (Step 5) are addressed only where they define the feed to, or the feed from, this operation; their characterization is out of scope. This plan governs execution only — the results are reported in the paired Process Characterization Report (**PCR-004**).

Basis. In the A-Mab case study, harvest and clarification has no product-quality impact: it defines the feed to Protein A and is characterized as a process-performance operation rather than a quality-forming one (CMC Biotech Working Group 2009). The Stage 1 approach and report structure follow PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the QbD framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012) and the lifecycle approach of the FDA 2011 guidance (U.S. Food and Drug Administration 2011).

2 Objectives

The characterization will:

1. confirm that harvest and clarification has **no impact on the product-quality CQAs**, which pass through the operation unchanged;
2. establish the effect of the operating parameters on the process-performance measures — product (step) yield, post-clarification turbidity, and depth-filter throughput;
3. confirm that a clarified feed of consistent quality (turbidity and impurity load) is delivered to the Protein A capture step across the operating ranges;
4. establish proven acceptable ranges (PARs) and confirm normal operating ranges (NORs) for each parameter; and
5. classify each parameter for its risk of impact to CQAs and to process performance (CPP / KPP / GPP) and provide the ranges that support the Stage 2 operating conditions and the drug-substance control strategy.

3 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 3.1: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-004	Process Characterization Report (Harvest and Clarification (Step 4))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3.2 (numbers are placeholders in this synthetic set).

Table 3.2: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Harvest and clarification separates the antibody-containing supernatant from cells and cell debris and defines the feed to Protein A. In the A-Mab platform the operation is a continuous disk-stack centrifugation followed by depth-filter and sterile-filter clarification; it removes cells and particulates but does not form, remove or modify any product-quality attribute (CMC Biotech Working Group 2009). Platform knowledge from related humanized IgG1 antibodies established that the glycan, charge-variant and aggregate attributes are set upstream in cell culture and that the host-cell protein (HCP) and DNA burden generated in culture is presented to, and cleared by, the purification train; harvest neither sets nor clears these attributes. This knowledge frames the process-performance measures in scope and the prioritization of parameters for characterization.

4.2 Quality attributes and process-performance measures

Harvest and clarification **sets or modifies no product-quality CQA**; the CQAs established in the production bioreactor pass through the operation unchanged. The operation is therefore characterized against the process-performance measures it governs (Table 4.1): the product (step) yield, the post-clarification turbidity that defines feed clarity to Protein A, and the depth-filter throughput that governs filter sizing and capacity. The impurity load presented to Protein A — the harvest HCP and residual DNA formed upstream — is monitored to confirm feed consistency but is not an attribute this step controls.

Table 4.1: Process-performance measures for the harvest and clarification characterization (no product-quality CQA is set by this step).

Process-performance measure	What it governs	Basis of the criterion
Product (step) yield	Antibody recovery across primary recovery	Mass balance; platform target
Post-clarification turbidity	Feed clarity presented to Protein A	NOR 1–10 NTU (Table 6.1)
Depth-filter throughput	Filter capacity / sizing and filtrate clarity	Depth-filter load range (Table 6.1)
Harvest HCP / residual-DNA load (monitored)	Consistency of feed to Protein A	Formed upstream; cleared downstream

4.3 Risk-based prioritization of parameters

Consistent with the A-Mab lifecycle of iterative risk assessments, the study scope is set by the Pre-Characterization Process Risk Assessment (**RA-001**). Because harvest and clarification carries no credible risk of impact to a product-quality CQA, none of its parameters was assigned to a multivariate design of experiments; each parameter is studied **one factor at a time** across its characterization range for its effect on the process-performance measures. The resulting parameter set and study-type assignment are given in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

Characterization will be performed on a qualified bench-scale clarification model that represents the commercial primary-recovery train — a laboratory disk-stack (or equivalent bench) centrifuge operated at the commercial equivalent relative centrifugal force and feed-solids loading, followed by scaled-down depth and sterile filters of the platform media at the commercial area-normalized load (**SOP-2005**, **SOP-2006**). Prior to characterization the model will be qualified against at-scale reference data (**SOP-1001**) by comparing centrate and filtrate turbidity, product yield and the HCP/DNA load presented to Protein A, confirming the model is representative and suitable to support the parameter ranges (Parenteral Drug Association 2013).

5.2 Operation

Harvest is initiated on the nominal duration criterion of the production culture. The culture is clarified by continuous centrifugation to remove cells and large debris, and the centrate is passed through a depth filter and a sterile (bioburden-reduction) filter to remove residual particulates before capture. The relative centrifugal force, the area-normalized depth-filter load and the resulting post-clarification turbidity are the operating variables; feed solids and culture duration are inputs governed by the upstream operation. Buffers, filters and media are controlled to platform specifications and are treated as identity/quality- controlled inputs rather than characterized variables (CMC Biotech Working Group 2009).

5.3 Analytical methods

Feed clarity and the feed impurity load will be measured by the validated methods in Table 5.1; method performance characteristics are per the cited validation reports.

Table 5.1: Validated analytical methods for the characterization.

Reference	Title	Type
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

5.4 Sampling plan

Each characterization run will be sampled at the clarified-pool stage for post-clarification turbidity and for the harvest HCP and residual-DNA load, and the product mass across the operation will be recorded for the step-yield calculation. The pressure/throughput profile of the depth filter will be recorded to confirm filter capacity. Runs at the parameter set-points will be replicated to quantify the reproducibility of yield and turbidity.

5.5 Statistical and analysis methods

Because no product-quality CQA is affected, the operation is characterized univariately rather than by a designed experiment (**SOP-1002** is not invoked for a multivariate design here). Each parameter will be varied one factor at a time across its characterization range, holding the others at their set-points, and the process-performance measures — step yield, post-clarification turbidity and depth-filter throughput — will be summarized against the parameter level. Reproducibility of yield and turbidity will be estimated from the replicated set-point runs. Parameter ranges will be judged acceptable where the performance measures remain within their platform targets and the product-quality CQAs are unchanged.

6 Study design

6.1 Parameters, ranges and study type

3 parameters are in scope (Table 6.1). The characterization range of each parameter ("Range studied") defines the edges of the explored knowledge space and the basis of the PAR; the NOR is the tighter routine-control range. Classification is an output of the study and is reported in PCR-004.

Table 6.1: Harvest / clarification parameters, set-points, characterization ranges and planned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Centrifugation (rcf)	g	9e+03	6000–12000	8000–10000	univariate
Depth filter load	L/m2	120	80–160	100–140	univariate
Post-clarification turbidity	NTU	5	1–20	1–10	univariate

6.2 Univariate ranging

Each parameter will be evaluated one factor at a time across its characterization range:

- **Centrifugation (relative centrifugal force)** — evaluated across its range for its effect on centrate clarity and product yield. Higher force clears the centrate more completely but increases the shear/lysis burden and the debris fines that load the depth filter; the range is bounded to protect both yield and feed clarity.
- **Depth-filter load (area-normalized)** — evaluated across its range for its effect on filtrate turbidity, filter capacity and yield. Higher load reduces filter area and cost but risks particulate breakthrough (higher turbidity) and pressure-driven yield loss near capacity.
- **Post-clarification turbidity** — evaluated as the controlled output attribute of the operation, confirming that the filtrate clarity presented to Protein A remains within its normal operating range across the operating conditions.

The product-quality CQAs will be confirmed unchanged across the ranging to demonstrate the absence of a product-quality impact.

7 Acceptance and decision criteria

The study is acceptable when:

1. the scale-down clarification model qualification confirms representativeness of the commercial primary-recovery train (**SOP-1001**);
2. the product-quality CQAs are demonstrated to be **unchanged** across the operation and across each parameter's characterization range; and
3. product yield, post-clarification turbidity and depth-filter throughput remain within their platform targets across each parameter's normal operating range, with reproducibility at the set-point sufficient to assure a consistent feed to Protein A.

Each parameter will then be classified per the A-Mab continuum (CMC Biotech Working Group 2009): a parameter that does not significantly impact a CQA but affects process performance or consistency is a **KPP**; a parameter with no meaningful impact on CQAs or performance over a wide range is a **GPP**. No harvest parameter is expected to be a CPP because the operation forms no CQA.

8 Data management and integrity

All raw data, analytical results and analyses will be recorded and retained under the site data-integrity procedures (ALCOA+). Analyses will be performed with version-controlled scripts and reviewed by a second analyst, and the ranging results will be archived with the report. Any deviation from this plan will be documented and assessed for impact in PCR-004.

9 Roles and responsibilities

Table 9.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the ranging study; author PCR-004
Analytical Development	Perform turbidity, HCP and DNA testing under the referenced AMV methods
Manufacturing Science	Qualify the scale-down clarification model; confirm representativeness
Purification Process Development	Confirm feed acceptability to Protein A capture
Quality Assurance	Approve the plan and the report; confirm parameter classifications

10 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report (**PCR-004**), reporting the executed ranging, the parameter classification and the confirmation that the operation has no product-quality impact. PCR-004 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment. Execution follows scale-down model qualification and precedes the Stage 2 PPQ campaign.

11 Risks and assumptions

The plan assumes the scale-down clarification model is representative of the commercial primary-recovery train (confirmed by qualification) and that the platform analytical methods are validated and in control. The principal risk is that the harvest impurity load or centrate/filtrate clarity shifts at commercial scale; this is mitigated by confirming the model at scale in Stage 2 and by carrying the KPP ranges forward with in-process monitoring of turbidity and the feed impurity load.

12 Approvals

Approval of this plan authorizes execution of the characterization described herein. Signatures are recorded in Table 1.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

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International Council for Harmonisation. 2023. *ICH Q9(R1): Quality Risk Management*. ICH.

International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.

Parenteral Drug Association. 2013. *Technical Report No. 60: Process Validation — a Lifecycle Approach*. PDA.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned univariate ranging

Each parameter is varied one factor at a time across its characterization range (Table 6.1) with the remaining parameters held at their set-points; the process-performance measures are recorded at each level.

Table 12.1: Appendix A — planned one-factor-at-a-time ranging design.

Parameter	Set-point	Range studied	NOR	Performance measures recorded
Centrifugation (rcf)	9e+03	6000–12000	8000–10000	step yield, post-clarification turbidity, depth-filter throughput
Depth filter load	120	80–160	100–140	step yield, post-clarification turbidity, depth-filter throughput
Post-clarification turbidity	5	1–20	1–10	step yield, post-clarification turbidity, depth-filter throughput